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Project Title: Distributed and Semi-distributed Machine Learning schemes over wireless environments

There is growing interest in distributed and semi-distributed machine learning techniques, which enable multiple edge devices to collaboratively train a model without revealing their raw data. These approaches offer the advantage of distributing computation across multiple devices while preserving the privacy of each device's data. Federated learning (FL), for instance, is a well-known semi-distributed learning method in which a central server coordinates with N edge devices to jointly train a model by solving an optimization problem. In this setup, data remains private to each device, and communication between the server and the devices is limited to the optimization parameters.

Meanwhile, recent years have seen increasing interest in learning over wireless environments, driven by the growing number of devices connected to servers via cellular networks. Given the massive amounts of data generated or accessed by mobile devices, both academia and industry are increasingly focusing on distributed and semi-distributed learning (e.g., FL) as a practical solution for training models on such data without needing to store it on a server. However, several challenges impede the implementation of this approach, particularly when devices and the server communicate over wireless channels, which suffer from communication and computation bottlenecks. For example, communication between devices and the server is subject to errors and attenuation/fading, which can degrade the accuracy and efficiency of the training process.

In this project, we will investigate communication-efficient training methods that reduce computational complexity and lower the communication overhead among devices. We will then analyze how wireless channels affect the performance of learning methods. Applications to fine-tuning LLMs will be considered.

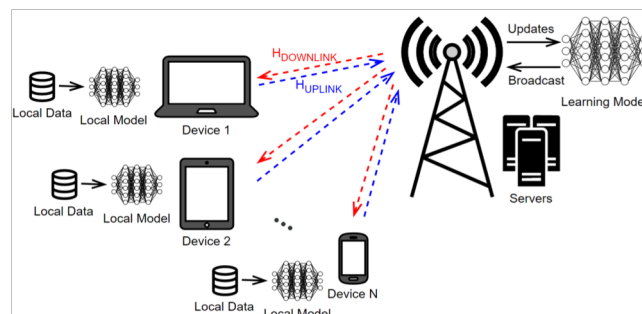


Figure 1: Federated learning over wireless networks.